



Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Fatima Jinnah women university



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OOP PROJECT

LIBRARY MANGEMENT SYSTEM

SOURCE CODE:

Login java:

```
package libraryms;
import java.awt.BorderLayout;
import java.awt.EventQueue;
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JLabel;
import javax.swing.SwingConstants;
import javax.swing.JTextField;
import javax.swing.JPasswordField;
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.Color;
import javax.swing.JButton;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;

public class login extends JFrame {

    private static final long serialVersionUID = 1L;
    private JPanel contentPane;
    private JTextField textField;
    private JPasswordField passwordField;

    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    login frame = new login();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }

    public login() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel_1_1 = new JLabel("username");
```

```

lblNewLabel_1_1.setBounds(115, 59, 68, 17);
lblNewLabel_1_1.setFont(new Font("Tahoma", Font.BOLD, 14));
contentPane.add(lblNewLabel_1_1);

textField = new JTextField();
textField.setBounds(233, 58, 162, 20);
textField.setBackground(new Color(255, 255, 153));
contentPane.add(textField);
textField.setColumns(10);

JLabel lblNewLabel = new JLabel("password");
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));
lblNewLabel.setBounds(115, 106, 68, 14);
contentPane.add(lblNewLabel);

passwordField = new JPasswordField();
passwordField.setBackground(new Color(255, 255, 153));
passwordField.setBounds(233, 105, 162, 20);
contentPane.add(passwordField);

JButton btnNewButton = new JButton("login");
btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
btnNewButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        openLibraryDashboard();
    }
});
btnNewButton.setBounds(111, 151, 89, 23);
contentPane.add(btnNewButton);

JButton btnNewButton_1 = new JButton("close");
btnNewButton_1.setFont(new Font("Tahoma", Font.BOLD, 14));
btnNewButton_1.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        System.exit(0);
    }
});
btnNewButton_1.setBounds(233, 151, 89, 23);
contentPane.add(btnNewButton_1);
}

protected void openLibraryDashboard() {
    librarymanagementsystem dashboard = new librarymanagementsystem ();
    dashboard.setVisible(true);
    this.dispose();
}
}

```

LIBRARYMANAGEMENTSYSTEM:

```
package libraryms;

import javax.swing.JButton;

import javax.swing.JFrame;

import javax.swing.JPanel;

import javax.swing.border.EmptyBorder;

import java.awt.EventQueue;

import java.awt.Font;

import java.awt.event.ActionEvent;

import java.awt.event.ActionListener;


public class librarymanagementsystem extends JFrame {


    private static final long serialVersionUID = 1L;

    private JPanel contentPane;


    /**
     * Launch the application.
     */
    public static void main(String[] args) {

        EventQueue.invokeLater(new Runnable() {

            public void run() {

                try {

                    librarymanagementsystem frame = new librarymanagementsystem();

                    frame.setVisible(true);
```

```
        } catch (Exception e) {  
            e.printStackTrace();  
        }  
    }  
});  
}
```

```
/**
```

```
 * Create the frame.
```

```
 */
```

```
public librarymanagementsystem() {  
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    setBounds(100, 100, 450, 300);  
    contentPane = new JPanel();  
    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));  
    setContentPane(contentPane);  
    contentPane.setLayout(null);  
  
    JButton btnNewButton = new JButton("BOOKS");  
    btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 12));  
    btnNewButton.setBounds(22, 11, 89, 23);  
    btnNewButton.addActionListener(new ActionListener() {  
        public void actionPerformed(ActionEvent e) {  
            // Open BooksFrame when "BOOKS" button is clicked
```

```
        BOOKS booksFrame = new BOOKS();

        booksFrame.setVisible(true);

    }

});

contentPane.add(btnNewButton);


JButton btnNewButton_1 = new JButton("PUBLISHER");

btnNewButton_1.setFont(new Font("Tahoma", Font.BOLD, 12));

btnNewButton_1.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open PublisherFrame when "PUBLISHER" button is clicked

        PUBLISHER publisherFrame = new PUBLISHER();

        publisherFrame.setVisible(true);

    }

});

btnNewButton_1.setBounds(121, 11, 89, 23);

contentPane.add(btnNewButton_1);


JButton btnNewButton_2 = new JButton("BORROWBY");

btnNewButton_2.setFont(new Font("Tahoma", Font.BOLD, 11));

btnNewButton_2.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open BorrowByFrame when "BORROWBY" button is clicked

        BORROWBBY borrowByFrame = new BORROWBBY();
```

```

        borrowByFrame.setVisible(true);

    }

});

btnNewButton_2.setBounds(223, 11, 89, 23);

contentPane.add(btnNewButton_2);


JButton btnNewButton_3 = new JButton("MEMBER");

btnNewButton_3.setFont(new Font("Tahoma", Font.BOLD, 12));

btnNewButton_3.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open MemberFrame when "MEMBER" button is clicked

        MEMBER memberFrame = new MEMBER();

        memberFrame.setVisible(true);

    }

});

btnNewButton_3.setBounds(322, 11, 89, 23);

contentPane.add(btnNewButton_3);


JButton btnNewButton_4 = new JButton("log out");

btnNewButton_4.setFont(new Font("Tahoma", Font.BOLD, 14));

btnNewButton_4.setBounds(162, 89, 89, 23);

contentPane.add(btnNewButton_4);

}

}

```

BOOKS JAVA:

```
package libraryms;
```

```
import java.awt.EventQueue;
```

```
import javax.swing.JFrame;
```

```
import javax.swing.JPanel;
```

```
import javax.swing.border.EmptyBorder;
```

```
import javax.swing.JLabel;
```

```
import javax.swing.JTextField;
```

```
import java.awt.Font;
```

```
import javax.swing.JButton;
```

```
public class BOOKS extends JFrame {
```

```
    private static final long serialVersionUID = 1L;
```

```
    private JPanel contentPane;
```

```
    private JTextField textField;
```

```
    private JTextField textField_1;
```

```
    private JTextField textField_2;
```

```
    private JTextField textField_3;
```

```
    private JTextField textField_4;
```

```
    private JTextField textField_5;
```

```
    /**
```

```
    * Launch the application.
```

```
    */
```

```
    public static void main(String[] args) {
```

```
        EventQueue.invokeLater(new Runnable() {
```

```

        public void run() {
            try {
                BOOKS frame = new BOOKS();
                frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    });
}

/**
 * Create the frame.
 */
public BOOKS() {
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setBounds(100, 100, 450, 300);
    contentPane = new JPanel();
    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

    setContentPane(contentPane);
    contentPane.setLayout(null);

    JLabel lblNewLabel = new JLabel("BOOK_ID:");
    lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));
    lblNewLabel.setBounds(62, 54, 91, 14);
    contentPane.add(lblNewLabel);

```

```
JLabel lblNewLabel_1 = new JLabel("AUTHOR:");  
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_1.setBounds(62, 79, 91, 14);  
contentPane.add(lblNewLabel_1);
```

```
JLabel lblNewLabel_2 = new JLabel("TITLE:");  
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_2.setBounds(62, 104, 91, 14);  
contentPane.add(lblNewLabel_2);
```

```
JLabel lblNewLabel_3 = new JLabel("PRICE:");  
lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_3.setBounds(62, 129, 91, 14);  
contentPane.add(lblNewLabel_3);
```

```
JLabel lblNewLabel_4 = new JLabel("AVAILABLE:");  
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_4.setBounds(62, 154, 91, 14);  
contentPane.add(lblNewLabel_4);
```

```
JLabel lblNewLabel_5 = new JLabel("PUBLISHER_ID:");  
lblNewLabel_5.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_5.setBounds(62, 179, 91, 14);  
contentPane.add(lblNewLabel_5);
```

```
textField = new JTextField();  
textField.setBounds(192, 53, 214, 20);  
contentPane.add(textField);
```

```
textField.setColumns(10);
```

```
textField_1 = new JTextField();  
textField_1.setBounds(192, 78, 214, 20);  
contentPane.add(textField_1);  
textField_1.setColumns(10);
```

```
textField_2 = new JTextField();  
textField_2.setBounds(192, 103, 214, 20);  
contentPane.add(textField_2);  
textField_2.setColumns(10);
```

```
textField_3 = new JTextField();  
textField_3.setBounds(192, 128, 214, 20);  
contentPane.add(textField_3);  
textField_3.setColumns(10);
```

```
textField_4 = new JTextField();  
textField_4.setBounds(192, 152, 214, 20);  
contentPane.add(textField_4);  
textField_4.setColumns(10);
```

```
textField_5 = new JTextField();  
textField_5.setBounds(192, 177, 214, 20);  
contentPane.add(textField_5);  
textField_5.setColumns(10);
```

```
JButton btnNewButton = new JButton("OK");
```

```
        btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
        btnNewButton.setBounds(126, 227, 89, 23);
        contentPane.add(btnNewButton);
    }

}
```

BORROWBBY:

```
package libraryms;
```

```
import java.awt.EventQueue;
```

```
import javax.swing.JFrame;
```

```
import javax.swing.JPanel;
```

```
import javax.swing.border.EmptyBorder;
```

```
import javax.swing.JLabel;
```

```
import javax.swing.JTextField;
```

```
import java.awt.Font;
```

```
import javax.swing.JButton;
```

```
public class BOOKS extends JFrame {
```

```
    private static final long serialVersionUID = 1L;
```

```
    private JPanel contentPane;
```

```
    private JTextField textField;
```

```
    private JTextField textField_1;
```

```
    private JTextField textField_2;
```

```
    private JTextField textField_3;
```

```
    private JTextField textField_4;
```

```

private JTextField textField_5;

/**
 * Launch the application.
 */
public static void main(String[] args) {
    EventQueue.invokeLater(new Runnable() {
        public void run() {
            try {
                BOOKS frame = new BOOKS();
                frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    });
}

/**
 * Create the frame.
 */
public BOOKS() {
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setBounds(100, 100, 450, 300);
    contentPane = new JPanel();
    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

    setContentPane(contentPane);

```

```
contentPane.setLayout(null);
```

```
JLabel lblNewLabel = new JLabel("BOOK_ID:");  
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel.setBounds(62, 54, 91, 14);  
contentPane.add(lblNewLabel);
```

```
JLabel lblNewLabel_1 = new JLabel("AUTHOR:");  
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_1.setBounds(62, 79, 91, 14);  
contentPane.add(lblNewLabel_1);
```

```
JLabel lblNewLabel_2 = new JLabel("TITLE:");  
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_2.setBounds(62, 104, 91, 14);  
contentPane.add(lblNewLabel_2);
```

```
JLabel lblNewLabel_3 = new JLabel("PRICE:");  
lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_3.setBounds(62, 129, 91, 14);  
contentPane.add(lblNewLabel_3);
```

```
JLabel lblNewLabel_4 = new JLabel("AVAILABLE:");  
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_4.setBounds(62, 154, 91, 14);  
contentPane.add(lblNewLabel_4);
```

```
JLabel lblNewLabel_5 = new JLabel("PUBLISHER_ID:");
```

```
lblNewLabel_5.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_5.setBounds(62, 179, 91, 14);  
contentPane.add(lblNewLabel_5);
```

```
textField = new JTextField();  
textField.setBounds(192, 53, 214, 20);  
contentPane.add(textField);  
textField.setColumns(10);
```

```
textField_1 = new JTextField();  
textField_1.setBounds(192, 78, 214, 20);  
contentPane.add(textField_1);  
textField_1.setColumns(10);
```

```
textField_2 = new JTextField();  
textField_2.setBounds(192, 103, 214, 20);  
contentPane.add(textField_2);  
textField_2.setColumns(10);
```

```
textField_3 = new JTextField();  
textField_3.setBounds(192, 128, 214, 20);  
contentPane.add(textField_3);  
textField_3.setColumns(10);
```

```
textField_4 = new JTextField();  
textField_4.setBounds(192, 152, 214, 20);  
contentPane.add(textField_4);  
textField_4.setColumns(10);
```

```
textField_5 = new JTextField();  
textField_5.setBounds(192, 177, 214, 20);  
contentPane.add(textField_5);  
textField_5.setColumns(10);
```

```
JButton btnNewButton = new JButton("OK");  
btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));  
btnNewButton.setBounds(126, 227, 89, 23);  
contentPane.add(btnNewButton);
```

```
}
```

```
}
```

MEMBER JAVA:

```
package libraryms;
```

```
import java.awt.EventQueue;
```

```
import javax.swing.JFrame;
```

```
import javax.swing.JPanel;
```

```
import javax.swing.border.EmptyBorder;
```

```
import javax.swing.JLabel;
```

```
import javax.swing.JTextField;
```

```
import java.awt.Font;
```

```
import javax.swing.JButton;
```

```
public class MEMBER extends JFrame {
```

```

private static final long serialVersionUID = 1L;
private JPanel contentPane;
private JTextField textField;
private JTextField textField_1;
private JTextField textField_2;
private JTextField textField_3;
private JTextField textField_4;

/**
 * Launch the application.
 */
public static void main(String[] args) {
    EventQueue.invokeLater(new Runnable() {
        public void run() {
            try {
                MEMBER frame = new MEMBER();
                frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    });
}

/**
 * Create the frame.
 */
public MEMBER() {

```

```
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
setBounds(100, 100, 450, 300);  
contentPane = new JPanel();  
contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
```

```
setContentPane(contentPane);  
contentPane.setLayout(null);
```

```
JLabel lblNewLabel = new JLabel("MEMBER_ID:");  
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel.setBounds(44, 66, 86, 14);  
contentPane.add(lblNewLabel);
```

```
JLabel lblNewLabel_1 = new JLabel("NAME:");  
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_1.setBounds(44, 91, 86, 14);  
contentPane.add(lblNewLabel_1);
```

```
JLabel lblNewLabel_2 = new JLabel("MEMEBR_DATE:");  
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_2.setBounds(44, 116, 86, 14);  
contentPane.add(lblNewLabel_2);
```

```
JLabel lblNewLabel_3 = new JLabel("EXPIRY_DATE:");  
lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_3.setBounds(44, 141, 86, 14);  
contentPane.add(lblNewLabel_3);
```

```
JLabel lblNewLabel_4 = new JLabel("ADDRESS:");  
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_4.setBounds(44, 166, 86, 14);  
contentPane.add(lblNewLabel_4);
```

```
textField = new JTextField();  
textField.setBounds(175, 63, 214, 20);  
contentPane.add(textField);  
textField.setColumns(10);
```

```
textField_1 = new JTextField();  
textField_1.setBounds(175, 88, 214, 20);  
contentPane.add(textField_1);  
textField_1.setColumns(10);
```

```
textField_2 = new JTextField();  
textField_2.setBounds(175, 113, 214, 20);  
contentPane.add(textField_2);  
textField_2.setColumns(10);
```

```
textField_3 = new JTextField();  
textField_3.setBounds(175, 138, 214, 20);  
contentPane.add(textField_3);  
textField_3.setColumns(10);
```

```
textField_4 = new JTextField();  
textField_4.setBounds(175, 163, 214, 20);  
contentPane.add(textField_4);
```

```
        textField_4.setColumns(10);

        JButton btnNewButton = new JButton("OK");
        btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
        btnNewButton.setBounds(120, 211, 89, 23);
        contentPane.add(btnNewButton);
    }

}
```

PUBLISHER JAVA:

```
package libraryms;

import java.awt.EventQueue;

import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JLabel;
import javax.swing.JTextField;
import java.awt.Font;
import javax.swing.JButton;

public class PUBLISHER extends JFrame {

    private static final long serialVersionUID = 1L;
    private JPanel contentPane;
    private JTextField textField;
    private JTextField textField_1;
```

```

private JTextField textField_2;

/**
 * Launch the application.
 */
public static void main(String[] args) {
    EventQueue.invokeLater(new Runnable() {
        public void run() {
            try {
                PUBLISHER frame = new PUBLISHER();
                frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    });
}

/**
 * Create the frame.
 */
public PUBLISHER() {
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setBounds(100, 100, 450, 300);
    contentPane = new JPanel();
    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

    setContentPane(contentPane);

```

```
contentPane.setLayout(null);
```

```
JLabel lblNewLabel = new JLabel("PUBLISHER_ID:");  
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel.setBounds(51, 74, 134, 14);  
contentPane.add(lblNewLabel);
```

```
JLabel lblNewLabel_1 = new JLabel("NAME:");  
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_1.setBounds(51, 113, 79, 14);  
contentPane.add(lblNewLabel_1);
```

```
JLabel lblNewLabel_2 = new JLabel("ADDRESS:");  
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_2.setBounds(51, 147, 79, 14);  
contentPane.add(lblNewLabel_2);
```

```
textField = new JTextField();  
textField.setBounds(199, 72, 186, 20);  
contentPane.add(textField);  
textField.setColumns(10);
```

```
textField_1 = new JTextField();  
textField_1.setBounds(199, 111, 186, 20);  
contentPane.add(textField_1);  
textField_1.setColumns(10);
```

```
textField_2 = new JTextField();
```

```
textField_2.setBounds(199, 145, 186, 20);
```

```
contentPane.add(textField_2);
```

```
textField_2.setColumns(10);
```

```
JButton btnNewButton = new JButton("OK");
```

```
btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
```

```
btnNewButton.setBounds(120, 193, 89, 23);
```

```
contentPane.add(btnNewButton);
```

```
}
```

```
}
```